



Ou Ocean Kun Hei
HKUST(GZ)
BSc in Data Science and Big Data Technology
(Expected May 2027)

+86 15112134690
✉ okhou533@connect.hkust-gz.edu.cn
🌐 <https://github.com/oukunhei>

EDUCATION

Hong Kong University of Science and Technology (Guangzhou)	2023 - 2027 (Expected)
Relevant Coursework: Theories in Computing, Deep Learning, Introduction to Natural Language Processing, Database Management Systems, Machine Learning, Discrete Mathematics, Applied Statistics	CGPA: 3.361/4.3

RESEARCH EXPERIENCE

DPC: Training-Free Text-to-SQL Candidate Selection via Dual-Paradigm Consistency (ACL2026)	X Program
We proposed the DPC (Dual-Paradigm Consistency) framework, use a multi-agent collaborative approach to synthesize Minimal Distinguishing Databases (MDD) , and leverage the complementarity of SQL and Python in execution logic to construct model-independent prior reference anchors. Achieving accuracy improvements of up to 2.2% over baselines like Self-Consistency.	2025.11 - 2026.1
As the second author, I was primarily responsible for conducting experiments across multiple models and systems, performing case analysis, and participating in paper revision and writing. The related work has been accepted by ACL 2026.	

INTERNSHIP

Lumotech	Internship
Lumotech is an AI-powered interior design platform for the B-end home furnishing industry that leverages LLM dialogue, 3D reconstruction, and AIGC to generate realistic preview renderings. As agent developer, I was responsible for the Agent module, designing custom skills and implementing tool-calling workflows to enable intuitive dialogue-driven interactions.	2026.5 - 2026.6

PROJECT EXPERIENCE

Large Model Self-Alignment Optimization System	Course Project
Reproduced a self-alignment pipeline for LLM post-training using OpenAssistant-Guanaco and LIMA datasets. Fine-tuned a Qwen2.5 reverse model with LoRA to generate instructions from responses and construct a synthetic instruction-tuning dataset. Applied LLM-based quality scoring and filtering to curate training samples, then performed supervised fine-tuning (SFT) of Qwen3-4B with LoRA, integrating data synthesis, quality assessment, and model adaptation into an end-to-end workflow .	2025.12
Data Science Project	Course Project
Developed a personalized study-report feature for an AnkiDroid-based Android application, processing learning performance data in Kotlin and integrating a Flask-hosted machine learning service. Applied Agile development practices through iterative releases, issue tracking, and collaborative code reviews. Automated APK builds and releases using GitHub Actions and implemented end-to-end tests for multiple-choice functionality. Authored technical documentation and project reports to document implementation decisions and development progress.	2025.9 - 2025.12

Multi-Tenant Kubernetes Platform

Built a multi-tenant Kubernetes lab platform on K3s, providing dedicated namespaces for independent teams sharing cluster infrastructure. Implemented namespace-scoped **RBAC** using Roles, RoleBindings, and ServiceAccounts for tenant administrators, developers, and viewers. Applied least-privilege permissions to separate workload management from sensitive operations involving secrets, access policies, and cluster administration. Enforced **resource governance** through ResourceQuota and LimitRange. Added **observability** through Prometheus and Grafana configuration. Integrated **Horizontal Pod Autoscaler (HPA)** to adjust workload replicas according to CPU utilization, configuring replica bounds and resource requests to support elastic scaling under variable demand within tenant resource constraints.

Course Project
2026.4 - 2026.5

SKILLS AND INTERESTS

Programming: Python, SQL, c++
Frameworks: PyTorch
Languages: English, Mandarin (native), Cantonese (native)
Areas of Interest: LLM Agent, Data Engineering, NL2SQL

VOLUNTEER ACTIVITIES

PLACE Summer Camp, HKUST(GZ)	2024, 2025
15th National Games of the People’s Republic of China	2025.11

HONORS & AWARDS

National Second Prize, 19th Chinese Collegiate Computing Design Contest (4c)	2026
Second-class Scholarship for Hong Kong, Macao, and Overseas Chinese Students	2025
Mirit Research Award of Undergraduate Research Program	2024
HKUST(GZ) Fisrt Class Scholarship	2023